Mustafa Abdel-Rahman Ibrahim Hassan

AI & Automation Intern | Petrochemical Processing Systems Engineering Graduate, Cairo University
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Summary

AI & Automation Intern and Petrochemical Processing Systems Engineering graduate (Cairo University, GPA 3.65) with hands-on experience building and shipping production AI agents. Skilled in designing LangGraph-based agent workflows, RAG pipelines, and n8n automations, and deploying them as secured, containerized FastAPI services on AWS. Combines a process-engineering foundation with applied machine learning, bringing a systems-thinking approach to agentic AI and workflow automation.

Education

2021 – Jun 2026 Cairo University, Faculty of Engineering – Credit Hours System (CHS)

Petrochemical Processing Systems Engineering (PPS) — Graduated
Cumulative GPA: **3.65**
Thesis: Design of a 1,500 ton/day industrial gypsum plant — mass and heat balances, equipment sizing, economic feasibility, and HAZID/HAZOP safety studies.

2018 – 2021 General Secondary Education Certificate

Experience

Teaching Assistant (Part-Time) – Cairo University, Faculty of Engineering | *Sep 2026 – Now*

AI Agents & Automation Intern – MIS | *Aug 2025 – Now*

- Designing and deploying **LangGraph agents** (PO Parser, Image Tagger) exposed as production services via **FastAPI**, triggered by Google Apps Script webhooks.
- Securing production endpoints with **HMAC authentication** and **SlowAPI** rate limiting.
- Deploying containerized agent services on **AWS EC2** with **Docker** and **Caddy**.
- Integrating **Supabase/Postgres** and **Airtable** for structured data storage and cross-system workflow automation.
- Building automated workflows with **n8n** for integrating and analyzing structured and unstructured data.

Social Media Intern – Waste Marche | *Jun 2025 – Jul 2025*

- Created LinkedIn carousel posts on sustainability, carbon footprint, waste management, and industrial training programs.
- Applied a data-driven content strategy to increase engagement and reach.

Process Engineer Intern – ENPPI | *Jul 2024*

- Gained in-depth knowledge of **towers, heat exchangers,** and process safety protocols.
- Learned industry best practices for safe and efficient process operations.

Process Engineer Intern – Onspec | *Aug 2024*

- Conducted **unsteady-state process simulation** in Aspen HYSYS.
- Studied **pump types, components, and performance calculations**.

Process Engineer Intern – Petrojet | *Jul 2023 – Aug 2023*

- Designed and interpreted **BFD, PFD, and P&ID** diagrams using AutoCAD.
- Collaborated on process data handling and documentation for engineering projects.

Process Engineer Intern – Onspec | *Aug 2023 – Sep 2023*

- Analyzed **System Flow Diagrams (SFD)** and utilities flow diagrams.
- Learned steady-state simulation techniques in Aspen HYSYS.

Skills

AI Agents & Automation: LangGraph, FastAPI, n8n, RAG pipelines, AI Agent & RAG Agent Development, Docker, AWS EC2, Caddy, Supabase, HMAC Auth, SlowAPI
Programming: Python, HTML/CSS
Data Science & Analytics: Pandas, NumPy, Scikit-learn, Data Cleaning, Feature Engineering, Data Visualization (Matplotlib, Seaborn), Statistical Analysis
Machine Learning: Regression, Classification, Clustering, Model Evaluation
Knowledge Graphs & Big Data: Neo4j / GDS (Leiden, KNN), GraphRAG architectures, Data Integration from Multiple Sources, Basic Cloud Computing Principles
Process Engineering Tools: Aspen HYSYS (steady state), Process Flow Diagrams (PFD), Piping & Instrumentation Diagrams (P&ID)
Other Tools: Microsoft tools, n8n Automation
Soft Skills: Analytical thinking, Problem-solving, Communication, Team Collaboration

Projects & Academic Work

AI Agent Automation Suite (MIS Production): Two LangGraph agents — a PO Parser and an Image Tagger — exposed via FastAPI and triggered by Google Apps Script webhooks; secured with HMAC auth and SlowAPI, and deployed on AWS EC2 using Docker and Caddy with

Supabase/Postgres and Airtable for data storage.

GraphRAG Knowledge Pipeline (Demo): Built a demo GraphRAG pipeline over a movie/actor dataset (IMDb, TMDB), comparing Microsoft GraphRAG, LightRAG, and LazyGraphRAG architectures; implemented entity extraction and Leiden community detection via Neo4j GDS in a dual-database design (Neo4j for graph traversal, Supabase for auth/metadata).

AI Automation Portfolio (Demo Projects): Developed a four-project freelance portfolio covering an AI image-tagging agent, a RAG chatbot workflow, n8n process-automation flows, and WhatsApp agent concepts, built with LangGraph, GPT-4o, FastAPI, Next.js, Supabase, and LangChain.

Chemical Process ML Model: Built a predictive Python model to optimize reaction yield using regression algorithms and preprocessing techniques.

Anomaly Detection in Multiphase Flow Facility (ML Project): Developed an ML-based anomaly detection system for a multiphase air–water flow facility using 29 sensor variables, performing data cleaning, feature engineering, and model comparison to identify normal vs. abnormal operation and classify two fault types (air leakage and air blockage).

Predicting Gas Estimated Ultimate Recovery (EUR) Using Machine Learning: Built a regression model to predict gas EUR for 507 simulated horizontal multistage fractured wells using geological, drilling, and completion attributes, achieving high correlation with actual EUR values while analyzing feature importance for reservoir performance forecasting.

Predicting Electrical Load Categories in Steel Production (DAEWOO Steel, South Korea): Developed a multi-class classification model to predict electricity load levels (Light, Medium, Maximum) for 35k+ steel production cycles using historical consumption data, KEPCO industry data, and time-based/grid features, enabling optimized production scheduling and load management.

Course And Workshops

- **Data Science Professional Certificate** – DataCamp (In Progress)
- **Circular Economy** – Cairo University (Feb 2024)
- **Process Safety** – ENPPI (Sep 2023)
- **Web Development** – Udacity (Sep 2022)

Language

Arabic: Native
English: Professional Proficiency

Interests

Building AI agents and automation systems, data-driven problem solving, and connecting Machine Learning with Process Engineering.